

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	1	of	18

NPS/002/019 – Technical Specification for LV Distribution and Service Cables

1. Purpose

The purpose of this document is to detail the requirements of CE Electric UK in relation to LV distribution and service cables for use on the company's distribution network

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/019	Version 2 (Issued Nov 08)	Specification for LV Distribution & Service Cables

2. Scope

This document details the technical requirements of CE Electric UK with respect to products detailed below and appendix 1.

- Single phase CNE and SNE service cables
- Three phase CNE and SNE service cables
- Waveform cables
- LV single core cables for sub-stations

3. Technical Requirements

3.1 Conditions of Installation.

Cables specified in this document shall be installed in accordance with the CE Electric UK policy for the installation of cables: DSS/026. Cables will be laid "direct" into the ground, pulled into buried ducts, open trenches or installed in air. During storage and after installation cables can be expected to be subjected to the full range of climatic conditions encountered in the UK.

Cable laid "direct", in ducts or trenches may be surrounded by ground water for most of their operating lives. Cables installed above ground will be supported by means of cleats either vertically or horizontally and may be exposed to direct sunlight for significant periods. Where installed on wood poles these cables may come into contact with a pole preservation medium such as creosote.

Cables installed on wood poles may require the removal of their oversheath materials to allow connections with other apparatus. Where this occurs it can result in possible long term exposure of the insulating materials covering the phase or neutral conductors. Manufacturers shall ensure that the polymers used to manufacture these materials will not suffer from UV degradation.

Accessories required to joint onto the cable may be cold applied or require the application of heat.

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	2	of	18

3.2 Conditions of Operation

Cables purchased in accordance with this specification are required to operate under conditions stipulated in IMP/001/911. The following are general conditions of operation:

- Nominal system voltages: 400 - 230 Volt.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 10%.
- Nominal system frequency: 50Hz
- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points.

3.3 Low Voltage Distribution Cables

3.3.1 3 & 4 Core Waveform Cable for CNE / SNE Systems

Three phase distribution cables for CNE systems shall comply with the requirements of BS 7870: Part 3, Section 3.40.

Phase conductors shall be sectoral shaped solid aluminium. The range of conductor sizes required is detailed in the appendix 1.

Phase insulation shall be XLPE type GP8 and complying with BS7655 – 1-3, however, the maximum permissible shrinkage shall be 2%.

Cores shall be laid up with a right hand direction of lay in the sequence Brown, Black, Grey (& Blue – for 4c cable).

The concentric Neutral/Earth wires shall be plain annealed copper bedded on a single rubber layer.

The oversheath shall be black PVC type TM1 and applied so that there is no internal protrusion of the oversheath material between or around the Neutral/Earth wires.

“LSOH” Cable as detailed in appendix 1 shall be in accordance with BS 7870-3.50

3.3.2 Single Phase Distribution Cables for Substation Connections

3.3.2.1 Cables with stranded copper conductor

Armoured cables shall comply with the requirements of BS 5467.

The range of conductor sizes and requirements for armoured versions are detailed in the accompanying schedules.

3.3.2.2 Cables with sectoral solid aluminium conductor

These cables shall comply with the requirements of BS 6346.

The range of conductor sizes and requirements for non-armoured versions are detailed in the accompanying schedules.

3.4 Low Voltage Service Cables

3.4.1 CNE Service Cables

Single-phase and three-phase CNE service cables shall comply with the requirements of BS 7870 part 3 section 3.11.

The conductor's shall be circular solid aluminium or circular stranded copper. Copper conductors may be compacted or uncompacted.

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	3	of	18

Phase insulation shall be XLPE type GP8 complying with BS7655 – 1-3, however, the maximum permissible shrinkage shall be 2%.

CNE Three phase service cables shall have the cores laid up with a right hand direction of lay in the sequence Brown, Black, Grey.

The over sheath shall be applied so that there is no internal protrusion of the over sheath material between or around the Neutral/Earth wires.

The range of conductor sizes and materials required is detailed in appendix 1.

“LSOH” Cable as detailed in appendix 1 shall be in accordance with BS 7870-3.12

3.4.2 SNE Service Cables

Single-phase and three phase split-concentric SNE service cables shall comply with the requirements of BS7870 part 3 section 3.21

The conductor's shall be circular solid aluminium or circular stranded copper. Copper conductors may be compacted or uncompacted.

Phase insulation shall be XLPE type GP8 complying with BS7655 – 1.3, however, the maximum permissible shrinkage shall be 2%.

Three phase service cables shall have the cores laid up with a right hand direction of lay in the sequence Brown, Black, and Grey.

The over sheath shall be applied so that there is no internal protrusion of the over sheath material between or around the Neutral and Earth wires.

The range of conductor sizes and materials required is detailed in appendix 1.

“LSOH” Cable as detailed in appendix 1 shall be in accordance with BS 7870-3.12

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	4	of	18

4. References

4.1 External Documentation

The cables shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant

4.2 British Standards

Reference	Title
BS 6234	Specification for polyethylene insulation and sheath of electric cables
BS EN 60228	Conductors in insulated cables
BS 7870-3.11	LV and MV Polymeric insulated cables for use by distribution and generation utilities – XLPE insulated combined neutral and earth copper wire concentric cables with copper or aluminium conductors
BS 7870-3.12	LV and MV Polymeric insulated cables for use by distribution and generation utilities – XLPE insulated combined neutral and earth copper wire concentric cables with copper or aluminium conductors having low emission of smoke and corrosive gases when effected by fire
BS 7870-3.21	LV and MV Polymeric insulated cables for use by distribution and generation utilities – XLPE insulated split concentric cables with copper or aluminium conductors
BS 7870-3.40	LV and MV Polymeric insulated cables for use by distribution and generation utilities – XLPE insulated, copper wire waveform concentric cables with solid aluminium conductors
BS 7655-1.3	Specification for Insulating and sheathing materials for cables – cross linked elastomeric insulating compounds

4.3 Internal documentation

Reference	Title
IMP/001/911	Code of Practise for the Economic Development of the LV System
DSS/026	Policy for the Installation of Distribution Power Cables

5 Definitions

Term	Definition
NIL	

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	5	of	18

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R. McMahon	Standards Engineer	R. McMahon	27/10/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M. Nicholson	System Strategy	M. Nicholson	21/10/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
V. Gorny	Standards Engineer	V. Gorny	05/10/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S. Johnson	DBD Administrator	S. Johnson	02/11/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C. Holdsworth	Standards and Assessment Manager	C. Holdsworth	27/10/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C. Morgan	Head of Procurement	C. Morgan	27/10/2009

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	6	of	18

Appendix 1 – Cable Types

3 - Core CNE Waveform Cables

- 95mm² Solid Aluminium Core & Copper Neutral Earth
- 185mm² Solid Aluminium Core & Copper Neutral Earth
- 300mm² Solid Aluminium Core & Copper Neutral Earth

4 - Core SNE Waveform Cables

- 95mm² Solid Aluminium Core & Copper Earth
- 185mm² Solid Aluminium Core & Copper Earth
- 300mm² Solid Aluminium Core & Copper Earth

3 - Core CNE Waveform Cables with LSOH Oversheath

- 185mm² Solid Aluminium Core & Copper Neutral Earth
- 300mm² Solid Aluminium Core & Copper Neutral Earth

4 - Core CNE Waveform Cables with LSOH Oversheath

- 95mm² Solid Aluminium Core & Copper Neutral Earth
- 185mm² Solid Aluminium Core & Copper Neutral Earth
- 300mm² Solid Aluminium Core & Copper Neutral Earth

Low Voltage Single Core Cables for Substations

- Armoured 800mm² Stranded Copper
- Un-armoured 480mm² Solid Sectoral Aluminium

Single Phase CNE Service Cables

- 16mm² stranded copper conductor
- 25mm² circular solid aluminium conductor
- 35mm² circular solid aluminium conductor
- 35mm² circular solid aluminium conductor **LSOH**

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	7	of	18

Three-Phase CNE Service Cables

- 35mm² circular solid aluminium conductors

Single Phase SNE Service Cables

- 16mm² stranded copper conductor
- 25mm² stranded copper conductor

Three- Phase SNE Service Cables

- 35mm² circular solid aluminium conductor

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	8	of	18

Appendix 2 – Addendum to Supplier Requirements

For each power cable offered the Tenderer shall provide the following cable rating data: -

- Cable rating laid direct (Amps)
- Cable rating laid in a 100mm internal diameter duct (Amps)
- Cable rating in air (Amps)

The oversheath of all cables shall be marked in accordance with the requirements of the relevant Standard specified in this document.

All cables shall be metre marked throughout the length of the cable and the start and end values shall be marked on the drum label.

Cables shall be marked with some form of reference or batch number that can be used to ensure batch trace ability of materials and manufacturing facilities used in the construction of the cable.

Tenderers shall submit a drawing showing the internal profile of the PVC sheath.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from the British standards or Energy Network Association Technical specifications.

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	9	of	18

Appendix 3 – Logistical Requirements

All service cable drums shall be non-returnable the tenderers shall state at the time of tender their proposed cable drum sizes and weights for each cable type offered.

All cable drums shall be marked in accordance with the relevant cable Specification or Standard. The drum label shall also contain

- (a) Customers catalogue number
- (b) Name of manufacturer
- (c) Supplied length
- (d) Rated voltage
- (e) Number of cores
- (f) Size of conductor
- (g) Type of conductor material ("Cu" or "Al")
- (h) Abbreviated description of cable construction
- (i) Gross and nett weights
- (j) Direction of rolling drum
- (k) The metre marking starts and end values
- (l) The unique reference number

Cable drums may be stored for long period's outdoors. All drum labels shall remain legible and durable under these conditions.

The ends of all cables shall be effectively sealed against the ingress of moisture by a method appropriate to the cable type. Tenderers shall detail at the time of tender their proposed sealing arrangement for each cable type offered.

The cable end projecting from the drum shall be protected from damage during transit, storage and handling on site.

The cable on the drum shall not be susceptible to damage during transit, storage and handling on site. Tenderers shall state at the time of tender their proposed method of protection for each cable

Each delivery length of cable shall be allocated a unique reference number. This number shall appear on the factory test sheet covering the cable length, shall be clearly marked on the drum on which the length is delivered and shall be referred to on all invoices and advice notes.

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	10	of	18

Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

LV distribution and service cables required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes, where appropriate indicating if tests are type or routine tests.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each type of cable offered and supplemented with a cable data sheet.

Document reference	NPS / 002 / 019						
Version:-	3	Date of Issue:-	Nov-09	Page	11	of	18

Specification for LV Distribution and Service Cables - 3Ø CNE / SNE Mains Cables				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Voltage Designation	BS 7870 3.4.3		
	Phase Conductor	BS 7870 3.4.4		
		BS 7870 3.4.14.2		
	Insulation	BS 7870 3.4.5		
		BS 7870 3.4.14.4		
		BS 7870 3.4.15.2		
		BS 7870 3.4.15.3		
	Assembly	BS 7870 3.4.6		
	Rubber Layer	BS 7870 3.4.7.3		
		BS 7870 3.4.14.5.1		
		BS 7870 3.4.14.5.2		
		BS 7870 3.4.15.4.1		
	Concentric Conductor	BS 7870 3.4.7		
		BS 7870 3.4.14.3		
	Oversheath	Generally to BS 7870 3.4.8 -Oversheath Material required Type TM1 to BS 7655 4.1		
		BS 7870 3.4.13.4		
		BS 7870 3.4.14.6		
		BS 7655 4.1 (Type TM 1)		
	Marking	BS 7870 3.4.9		
		BS 7870 3.4.13.5		
	End Sealing	BS 7870 3.4.10		
	Dimensions	BS 7870 3.4.11 Tables 1 (CNE)		
		BS 7870 3.4.11 Tables 2 (SNE)		

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	12	of	18

Specification for LV Distribution and Service Cables - 3Ø CNE / SNE Mains Cables				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Tests:	BS 7870 3.4.12 Table 3		
	Conductor Resistance	BS 7870 3.4.13.2		
	Voltage Test	BS 7870 3.4.13.3		
	Spark Tests	BS 7870 3.4.13.4		
	Compatibility	BS 7870 3.4.15.6 Table 4		
	LSOH Option	Generally to BS 7570 3.5		

Document reference	NPS / 002 / 019				
Version:-	3	Date of Issue:-	Nov-09	Page	13 of 18

Specification for LV Distribution and Service Cables - CNE Service Cables				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Voltage Designation	BS 7870 3.1.3		
	Phase Conductor	BS 7870 3.1.4		
		BS 7870 3.1.15.2		
	Insulation	Generally to BS 7870 3.1.5 - Insulation Material required GP 8 BS 7655 1.3		
		BS 7870 3.1.14.4		
		BS 7870 3.1.15.4		
	Assembly	BS 7870 3.1.6		
	Bedding	BS 7870 3.1.7		
	Concentric Conductor	BS 7870 3.1.8		
		BS 7870 3.1.15.3		
	Oversheath	BS 7870 3.1.9		
		BS 7870 3.1.15.5		
		BS 7870 3.1.16.3		
	Marking	BS 7870 3.1.10		
		BS 7870 3.1.14.6		
	End Sealing	BS 7870 3.1.11		
	Flame Propagation	BS 7870 3.1.15.6		
	Dimensions	BS 7870 3.1.12 Table 1 (1Ø Cu)		
		BS 7870 3.1.12 Table 2 (1Ø Al)		
		BS 7870 3.1.12 Table 3 (3Ø Cu)		
		BS 7870 3.1.12 Table 4 (3Ø Al)		

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	14	of	18

Specification for LV Distribution and Service Cables - CNE Service Cables				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Tests:	BS 7870 3.1.13 Table 5		
	Conductor Resistance	BS 7870 3.1.14.2		
	Voltage Test	BS 7870 3.1.14.3		
	Insulation Resistance	BS 7870 3.1. 14.4 Table 6		
	Spark Tests	BS 7870 3.1 14.5		
	Cable Markings	BS 7870 3.1 14.6		
	General	BS 7870 3.1.15.1 / 3.1.16.1 Table 5		
	Compatibility	BS 7870 3.1.16.4 Table 7		
	LSOH Option	Generally to BS 7870-3.12		

Document reference	NPS / 002 / 019						
Version:-	3	Date of Issue:-	Nov-09	Page	15	of	18

Specification for LV Distribution and Service Cables - SNE Service Cables				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Voltage Designation	BS 7870 3.2.3		
	Phase Conductor	BS 7870 3.2.4		
		BS 7870 3.2.14.2		
	Insulation	Generally to BS 7870 3.2.5 - Insulation Material required GP 8 BS 7655 1.3		
		BS 7870 3.2.13.4		
		BS 7870 3.2.13.5		
		BS 7870 3.2.14.4		
		BS 7870 3.4.15.2 (Type GP8)		
	Concentric Conductors	BS 7870 3.2.6.1 / 3.2.6.2		
		BS 7870 3.2.14.3		
		BS 7870 3.2.15.3 (neutral)		
	Separators	BS 7870 3.2.6.3		
	Application	BS 7870 3.2.6.4		
		BS 7870 3.2.14.3		
	Binders	BS 7870 3.2.7		
	Oversheath	BS 7870 3.2.8		
		BS 7870 3.2.13.5		
		BS 7870 3.2.14.5		
		BS 7870 3.2.15.4		
	Marking	BS 7870 3.2.9		
		BS 7870 3.2.13.6		
	End Sealing	BS 7870 3.2.10		
	Flame Propagation	BS 7870 3.2.14.6		

Document reference		NPS / 002 / 019					
Version:-	3	Date of Issue:-	Nov-09	Page	16	of	18

Specification for LV Distribution and Service Cables - SNE Service Cables				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Dimensions	BS 7870 3.2.11 Table 1 (1Ø Cu)		
		BS 7870 3.1.11 Table 2 (1Ø Al)		
	Tests:	BS 7870 3.2.12 Table 3		
	Conductor Resistance	BS 7870 3.2 13.2		
	Voltage Test	BS 7870 3.2.13.3		
	Insulation Resistance	BS 7870 3.2. 13.4 Table 4		
	Spark Tests	BS 7870 3.2 13.5		
	Cable Markings	BS 7870 3.2 13.6		
	General	BS 7870 3.2.13.1 / 3.2.14.1 Table 3		
	Compatibility	BS 7870 3.2.15.5 Table 5		
	LSOH Option	Generally to BS 7870-3.12		

Document reference	NPS / 002 / 019				
Version:-	3	Date of Issue:-	Nov-09	Page	17 of 18

Specification for LV Distribution and Service Cables – Substation 1c Cables (Stranded Cu)				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Voltage Designation	BS 5467 4 (600/1000v)		
	Phase Conductor	BS 5467 5		
	Insulation:			
	Material	BS 5467 6.1 Cross-linked Polyethylene (GP8) to BS 7655 1.3		
	Thickness	BS 5467 6.2		
	Testing	BS 5467 6.3		
	Identification	BS 5467 7		
	Bedding	BS 5467 9.1		
		BS 5467 9.2		
	Armours	BS 5467 10.3		
		BS 5467 10.5		
	Oversheath:	BS 5467 11		
	General	BS 5467 11.1		
	Thickness	BS 5467 11.2		
	Testing	BS 5467 11.3		
	Marking	BS 5467 12		
	End Sealing	BS 5467 13		
	Flame Propagation	BS 5467 17.2		
	Dimensions	BS 5467 Table 4 (Stranded Cu)		
	Tests:	BS 5467 Table 2 (Schedule of Tests)		
	Conductor Resistance	BS 5467 16.2		
	Voltage Test	BS 5467 16.3		
	Insulation Shrinkage	BS 5467 18.3		
	Compatibility	BS 5467 Table 3		
		BS 5467 18.2		

Document reference	NPS / 002 / 019						
Version:-	3	Date of Issue:-	Nov-09	Page	18	of	18

Specification for LV Distribution and Service Cables – Substation 1c Cables (Solid Sectoral Al)				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Voltage Designation	BS 6346 4 (600/1000v)		
	Phase Conductor	BS 6346 5		
	Insulation:			
	Material	BS 6346 6.1		
	Thickness	BS 6346 6.2		
	Testing	BS 6346 6.3		
	Identification	BS 6346 7		
	Bedding	BS 6346 9.1		
		BS 6346 9.2		
	Oversheath:			
	General	BS 6346 11.1		
	Thickness	BS 6346 11.2		
	Testing	BS 6346 11.3		
	Marking	BS 6346 12		
	End Sealing	BS 6346 13		
	Flame Propagation	BS 6346 17.2		
	Dimensions	BS 6346 Table 6 (Solid Al)		
	Tests:	BS 6346 Table 2 (Schedule of Tests)		
	Conductor Resistance	BS 6346 16.2		
	Voltage Test	BS 6346 16.3		
	Insulation Resistance	BS 6346 16.4		
	Compatibility	BS 6346 Table 4		
		BS 6346 18.2		